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ORIGINAL PAPER

Approximate Solution and Analysis of Smoking Epidemic Model with Caputo Fractional Derivatives

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Abstract
In this paper, an epidemic model is presented to describe the dynamics of drugs usage among the adults. The Caputo fractional derivative operator of order $\phi \in (0, 1]$ is employed to obtain the system of fractional differential equations. Smoking is the large problem in the entire world. Despite the overwhelming facts about smoking, it is still a bad habit which is widely spread and socially accepted. A non-linear mathematical model is established to study and assess the dynamics of smoking and its impact on public health in a community. The analysis of two different states, disease free and endemic which means the disease dies out or persist in a population has been made by presenting sensitivity analysis. The stability of the model is provided by threshold or reproductive number. The Laplace Adomian decomposition method has been employed to solve smoking model which has five compartments, potential smoker P, occasional smokers O, smokers S, temporarily quit smokers Q and permanently quit smokers L. Finally, some numerical results are presented which shows the effect of fractional parameter ϕ on our obtained solutions.

Keywords Epidemic model · Endemic equilibrium · Qualitative analysis · Stability analysis · Fractional order

Introduction

The scope of mathematics includes mathematical modeling and esoteric mathematics. The flow of work, process, predictions and outcomes can easily be measured with the help of mathematical concepts and theory. Therefore, biologists are now extremely dependent on mathematics. Mathematical modeling of biological sciences is presented by many brilliant scientist [1–3]. The mathematical modeling involves biological system, integer order differential equations that show their dynamics and complex system which describes their structure. The nonlinearity and multi-scale behaviors in mathematical modeling describe the

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mutual relationship between parameters [4]. In last few decades, many biological models were studied in detail by using classical derivatives some of the studies are [5–9].

Smoking is the major problem in the entire world effecting healthy community. Smoking effects different organs of human body caused more than one million deaths in the world. A chance of heart attack in smoker is 70% more as compared to nonsmoker. Similarly, the incident rate of lung cancer of smoker is 10% more than nonsmoker. The main effects of short term smoking are coughing, stained teeth, high blood pressure and bad breath. The major effects of long term smoking are gum disease, stomach ulcer, lung cancer, heart disease, throat cancer and mouth cancer in the recent years. The life of smoker is also 12–13 years shorter than non-smoker. According to the reports of world health organization (WHO) smoking kills many individual in the entire world. Every scientist, doctor and mathematician tries to control the effect of smoking. The lot of smoking models are presented by the researchers. Erturk et al. examined the smoking model related with Caputo fractional derivative [10]. Zaman studied the optimal control of the smoking models and present the qualitative analysis of dynamics of smoking [11,12]. Analysis of cigarette smoking and lung cancer is presented in [13]. Garsow et al. [14] describe the mathematical analysis of tobacco use their decline. Some more interesting studies about dynamics of smoking is discussed in [15–18].

A generalization of ordinary differentiation and integration to complex and non- integer order were presented in fractional calculus [19]. In recent years, considerable interest in fractional differential equations has been stimulated due to their numerous applications in the areas of physics and engineering. Many important phenomena are well described by fractional differential equations in electro magnetics, acoustics, visco-elasticity, electrochemistry and material science. That is because of the fact that a realistic modeling of a physical phenomenon having dependence not only at the time instant, but also the previous time history can be successfully achieved by using fractional calculus [20–23]. The basic concepts of fractional differential equations and their applications are given in [24]. Bulut et al. [25] studied the differential equations of arbitrary order by applying analytical techniques. In the branch of fractional calculus, fractional derivatives and fractional integrals are important aspects. Recently, many researchers and scientists have analyzed this special branch. Caputo introduced a fractional derivative which allows the conventional initial and boundary conditions associated with the real world problem. Many researcher reported new advances in fractional calculus and related issues in their monographs [26–29].

The coupling of ADM and Laplace transform leads to a powerful method known as Laplace Adomain decomposition method (LADM). With the help of Laplace transform, we convert a differential equations to an algebraic equations and the nonlinear terms are decomposed in terms of Adomain polynomials. This analytical method works efficiently for a system of deterministic as well as stochastic differential equations. More explicitly, it can be used for a system of linear and nonlinear ordinary and partial differential equations of classical as well as fractional order. In this method no perturbation or liberalization is required. Further it has no need of pre-defined step size like RK4. It is to be noted that LADM is powerful than standard ADM method [30–32].

The major objective of this work is to apply the new fractional derivative in smoking model and communicating the particulars of the exactness and uniqueness of the solution by using the fixed point theorem. The is organized as follows. In Sect. 2 materials and method is given. In Sect. 3, stability is discussed. In Sect. 4, some preliminaries related to CF (Caputo Fractional) derivatives is discussed. In Sect. 5, solution of fractional smoking model is presented. Results and discussion are given in Sect. 6. Lastly in Sect. 7, the concluding remarks are presented.

Materials and Method

The concept of mathematical modeling has been prolonged to define the stability and qualitative features of giving up smoking models from 2000 [11,12]. Smoking model is divided into five subdepartment like potential smoker $P(t)$, the occasional smoker $O(t)$, smoker $S(t)$, temporarily quit smoker $Q(t)$ and permanently quit smoker $L(t)$. The $N(t)$ is denotes the total population size at time t such that $N(t) = P(t) + O(t) + S(t) + Q(t) + L(t)$. The proposed smoking model [33] in the form of system of nonlinear differential equation is given as:

$$\frac{dP}{dt} = \Lambda - \beta PS - \mu P, \tag{1}$$

$$\frac{dO}{dt} = \beta PS - \alpha_1 O - \mu O, \tag{2}$$

$$\frac{dS}{dt} = \alpha_1 O + \alpha_2 SQ - (\mu + \gamma)S, \tag{3}$$

$$\frac{dQ}{dt} = -\alpha_2 SQ - \mu Q + \gamma(1 - \delta)S, \tag{4}$$

$$\frac{dL}{dt} = \delta\gamma S - \mu L, \tag{5}$$

with initial conditions $P(0) = n_1$, $O(0) = n_2$, $S(0) = n_3$, $Q(0) = n_4$, $L(0) = n_5$, where Λ is represent the recruitment rate in P, β is the effective contact rate between S and P, μ is the natural death rate, α_1 is the rate at which occasional smokers become regular smokers, α_2 and γ represents the contact rate between smokers and temporary quitters who revert back to smoking and the rate of quitting smoking respectively, $(1 - \delta)$ is the fraction of smokers who temporary quit smoking (at the rate γ), δ is the remaining fraction of smoking which represents permanently quit smoking [33].

The fractional order extension of this model have been first studied in [15,16]. The reason of using fractional differential equations (FDEs) is that to show the realistic biphasic decline behavior of infection of diseases but at a slower rate. The system of fractional differential equations (FDEs) describes as follows.

$$\begin{cases} D^{\phi_1} P(t) = \Lambda - \beta PS - \mu P, \\ D^{\phi_2} O(t) = \beta PS - \alpha_1 O - \mu O, \\ D^{\phi_3} S(t) = \alpha_1 O + \alpha_2 SQ - (\mu + \gamma)S, \\ D^{\phi_4} Q(t) = -\alpha_2 SQ - \mu Q + \gamma(1 - \delta)S, \\ D^{\phi_5} L(t) = \delta\gamma S - \mu L, \end{cases} \tag{6}$$

with initial conditions

$$P(0) = n_1,\ O(0) = n_2,\ S(0) = n_3,\ Q(0) = n_4,\ L(0) = n_5. \tag{7}$$

The system is qualitatively analyzed by two ways i.e. disease free equilibrium and endemic equilibrium.

Equilibrium Point and Stability

For equilibrium point, we take

$$D^{\phi_1} P(t) = D^{\phi_2} O(t) = D^{\phi_3} S(t) = D^{\phi_4} Q(t) = D^{\phi_5} L(t) = 0, \tag{8}$$

in system (6), we get disease free equilibria as

$$E_0 = (P, 0, S, 0, 0),$$

and endemic equilibria of the system is

$$E^* = (P^*, P^*, S^*, Q^*, L^*),$$

where

$$P^* = \frac{\Lambda}{\beta S + \mu e}, \quad O^* = \frac{\beta \Lambda S}{(\beta S + \mu)(\alpha_1 + \mu)}, \quad Q^* = \frac{\gamma(1-\delta)S}{\alpha_2 S + \mu}, \quad L^* = \frac{\delta \gamma S}{\mu}. \tag{9}$$

Theorem 1 *The disease-free equilibrium E^* is locally asymptotically stable if $R_0 < 1$ and is unstable if $R_0 > 1$.*

Reproductive Number

It is important to find the verge conditions of model to check the status of population, whether the disease persist or dies out. The reproductive number R_0 in epidemiological model is defined as follows:

In case of disease free equilibrium, $R_0 < 1$, which shows that the disease will dies out.

In case of endemic equilibrium, $R_0 > 1$, which shows that the disease spreads in the population.

The reproductive number is also known basic reproductive ratio or basic reproductive rate.

Proposition 1 *The disease free equilibrium $E.$ is locally asymptotically stable for $R_0 < 1$, otherwise unstable. Consider the jacobian matrix as*

$$J = \begin{bmatrix} -\beta S - \mu & 0 & -\beta P & 0 & 0 \\ \beta S & -\alpha_1 - \mu & \beta P & 0 & 0 \\ 0 & \alpha_1 & \alpha_2 Q - \gamma - \mu & 0 & 0 \\ 0 & 0 & -\alpha_2 Q + \gamma(1-\delta) & -\alpha_2 S - \mu & 0 \\ 0 & 0 & \gamma\delta & 0 & -\mu \end{bmatrix}. \tag{10}$$

Since the Jacobian matrix is $J = F - V$ then the matrix F and V can be written as

$$F = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & \alpha_1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \tag{11}$$

$$V = \begin{bmatrix} \beta S + \mu & 0 & \beta P & 0 & 0 \\ -\beta S & \alpha_1 + \mu & -\beta P & 0 & 0 \\ 0 & 0 & -\alpha_2 Q + \gamma + \mu & 0 & 0 \\ 0 & 0 & \alpha_2 Q - \gamma(1-\delta) & \alpha_2 S + \mu & 0 \\ 0 & 0 & -\gamma\delta & 0 & \mu \end{bmatrix}. \tag{12}$$

We know that $K = FV^{-1}$ and using the relation $|K - \lambda I| = 0$ for the eigen value λ , we get

$$\lambda = \frac{\alpha_1 \beta \mu P}{(\alpha_1 + \mu)(\mu + \gamma)(\mu + \beta S)}.$$

by substituting the values of each parameters, we get $0 < \lambda < 1$, which shows the reproductive number $R_0 < 1$. Since reproductive number $R_0 < 1$, so the constructed system is in disease free state. Reproductive number in this system (6) is

$$R_0 = \frac{\alpha_1 \beta \mu P}{(\alpha_1 + \mu)(\mu + \gamma)(\mu + \beta S)}.$$

We see that all the eigen values are negative only for $R_0 < 1$ so this is in disease free state is locally asymptotically stable for $R_0 < 1$, otherwise unstable.

Sensitivity Analysis of R_0

The sensitivity of R_0 to each of its parameters is

$$\begin{aligned}
\frac{\partial R_0}{\partial \alpha_1} &= \frac{\beta \mu^2 P}{(\alpha_1 + \mu)^2(\mu + \gamma)(\mu + \beta S)} \geq 0,\\
\frac{\partial R_0}{\partial \beta} &= \frac{\alpha_1 \mu^2 P}{(\alpha_1 + \mu)(\mu + \gamma)(\mu + \beta S)^2} \geq 0,\\
\frac{\partial R_0}{\partial \gamma} &= -\frac{\alpha_1 \beta \mu^2 P}{(\alpha_1 + \mu)(\mu + \gamma)^2(\mu + \beta S)} \leq 0,\\
\frac{\partial R_0}{\partial \mu} &= \frac{\beta \mu^2 P[(\alpha_1 + \mu)^2(\mu + \gamma)(\mu + \beta S) - \mu]}{[(\alpha_1 + \mu)(\mu + \gamma)(\mu + \beta S)]^2} \geq 0.
\end{aligned}$$

It can be seen that R_0 is most sensitive to change in parameter, here, R_0 is increasing with $\alpha_1,\ \beta,\ \mu$ and decreasing with γ. In other words it found that the sensitivity analysis shows that prevention is better than quit smoking.
Non negative solution Let $R^5_+ = \{x \in R^5, x \geq 0\}$ and $x(t) = (P(t), O(t), S(t), Q(t), L(t))^T$.

Lemma (Generalized mean value theorem) *Let* $h(x) \in C[a, b]$ *and* $D^\phi h(x) \in C[a, b]$ *for* $0 < \phi \leq 1$. *Then, we have*

$$h(x) = h(a) + \frac{1}{(\phi + 1)!} D^\phi h(\eta)(x - a)^\phi. \tag{13}$$

with $0 \leq \eta \leq x$ *for all* $x \in (a, b]$.

Theorem 2 *There is a unique solution for the initial value problem given in system* (6)*, and the solution remains in* $R^5, x \geq 0\}$.

Proof The existence and uniqueness of the solution of system (6), in $(0, \phi)$. Our aim is proof that the domain $R^5, x \geq 0$ is positively invariant.

$$D^\phi P(t)|_{P=0} = \Lambda - \mu P \geq 0, \tag{14}$$
$$D^\phi O(t)|_{O=0} = \beta P S \geq 0, \tag{15}$$
$$D^\phi S(t)|_{S=0} = \alpha_1 O \geq 0, \tag{16}$$
$$D^\phi Q(t)|_{Q=0} = \gamma(1 - \delta) S \geq 0, \tag{17}$$
$$D^\phi L(t)|_{L=0} = \delta \gamma S \geq 0. \tag{18}$$

On each hyper-plane bounding the nonnegative orthant, the vector field points into R^+_5. □

Preliminaries

In this section, we give some fundamental results and definitions from fractional calculus. For detailed over view of the topic readers are referred to [34–36].

Definition 4.1 The Riemann–Liouville fractional integration of order $\alpha \in (0, 1)$ of the function $f \in L^1([0, T], R)$ is defined as as

$$I_{0+}^{\alpha} f(t) = \frac{1}{\Gamma(\alpha)} \int_0^t (t-s)^{\alpha-1} f(s) ds.$$

The Riemann–Liouville derivative has certain disadvantages such that the fractional derivative of a constant is not zero. Therefore, we will make use of Caputo's definition owing to its convenience for initial conditions of the fractional differential equations.

Definition 4.2 The Caputo fractional order derivative of a function f on the interval $[0, T]$ is defined by

$${}^{c}D_{0+}^{\alpha} f(t) = \frac{1}{\Gamma(n-\alpha)} \int_0^t (t-s)^{n-\alpha-1} f^{(n)}(s) ds,$$

where $n = [\alpha] + 1$ and $[\alpha]$ represents the integer part of α. In particularly, for $0 < \alpha < 1$, Caputo derivatives becomes

$${}^{c}D_{0+}^{\alpha} f(t) = \frac{1}{\Gamma(1-\alpha)} \int_0^t \frac{f^{'}(s)}{(t-s)^{\alpha}} ds,$$

Lemma 4.1 *The following results holds for fractional differential equations*

$$I^{\alpha}[{}^{c}D^{\alpha} h](t) = h(t) + \sum_{i=0}^{n-1} \frac{h^{(i)}}{i!} t^i,$$

for arbitrary $\alpha > 0$, $i = 0, 1, 2, \ldots, n-1$, *where* $n = [\alpha] + 1$ *and* $[\alpha]$ *represents the integer part of* α.

Definition 4.3 The definitions of Laplace transform of Caputo's derivative is written as

$$L\{{}^{c}D^{\alpha} y(t)\} = s^{\alpha} h(s) - \sum_{k=0}^{n-1} s^{\alpha-i-1} y^{(k)}(0), \quad n-1 < \alpha \leq n; \text{‘ } n \in N.$$

for arbitrary $c_i \in R$, $i = 0, 1, 2, \ldots, n-1$, where $n = [\alpha] + 1$ and $[\alpha]$ represents the integer part of α.

The Laplace–Adomian Decomposition Method

Consider the fractional-order epidemic model (6) subject to the initial condition (7). The nonlinear term in this model is PS and SQ. For $\phi_1 = \phi_2 = \phi_3 = \phi_4 = \phi_5 = 1$ the fractional order model converts to the classical epidemic model. By using the Laplace transform on the system (6), we get

$$\mathcal{L}\{D^{\phi_1} P(t)\} = \Lambda \mathcal{L}\{1\} - \beta \mathcal{L}\{PS\} - \mu \mathcal{L}\{P\}, \tag{19}$$

$$\mathcal{L}\{D^{\phi_2} O(t)\} = \beta \mathcal{L}\{PS\} - (\alpha_1 + \mu) \mathcal{L}\{O\}, \tag{20}$$

$$\mathcal{L}\{D^{\phi_3}S(t)\} = \alpha_1\mathcal{L}\{O\} + \alpha_2\mathcal{L}\{SQ\} - (\mu+\gamma)\mathcal{L}\{S\}, \tag{21}$$
$$\mathcal{L}\{D^{\phi_4}Q(t)\} = -\alpha_2\mathcal{L}\{SQ\} - \mu\mathcal{L}\{Q\} + \gamma(1-\delta)\mathcal{L}\{S\}, \tag{22}$$
$$\mathcal{L}\{D^{\phi_5}L(t)\} = \delta\gamma\mathcal{L}\{S\} - \mu\mathcal{L}\{L\}, \tag{23}$$

by applying the property of Laplace transform, we get the as follows

$$S^{\phi_1}\mathcal{L}\{P\} - S^{\phi_1-1}P(0) = \Lambda\mathcal{L}\{1\} - \beta\mathcal{L}\{PS\} - \mu\mathcal{L}\{P\}, \tag{24}$$
$$S^{\phi_2}\mathcal{L}\{O\} - S^{\phi_2-1}O(0) = \beta\mathcal{L}\{PS\} - (\alpha_1+\mu)\mathcal{L}\{O\}, \tag{25}$$
$$S^{\phi_3}\mathcal{L}\{S\} - S^{\phi_3-1}S(0) = \alpha_1\mathcal{L}\{O\} + \alpha_2\mathcal{L}\{SQ\} - (\mu+\gamma)\mathcal{L}\{S\}, \tag{26}$$
$$S^{\phi_4}\mathcal{L}\{Q\} - S^{\phi_4-1}Q(0) = -\alpha_2\mathcal{L}\{SQ\} - \mu\mathcal{L}\{Q\} + \gamma(1-\delta)\mathcal{L}\{S\}, \tag{27}$$
$$S^{\phi_5}\mathcal{L}\{L\} - S^{\phi_5-1}L(0) = \delta\gamma\mathcal{L}\{S\} - \mu\mathcal{L}\{L\}, \tag{28}$$

by rearranging the above equations, we have

$$S^{\phi_1}\mathcal{L}\{P\} = S^{\phi_1-1}P(0) + \Lambda\mathcal{L}\{1\}\beta\mathcal{L}\{PS\} - \mu\mathcal{L}\{P\}, \tag{29}$$
$$S^{\phi_2}\mathcal{L}\{O\} = S^{\phi_2-1}O(0) + \beta\mathcal{L}\{PS\} - (\alpha_1+\mu)\mathcal{L}\{O\}, \tag{30}$$
$$S^{\phi_3}\mathcal{L}\{S\} = S^{\phi_3-1}S(0) + \alpha_1\mathcal{L}\{O\} + \alpha_2\mathcal{L}\{SQ\} - (\mu+\gamma)\mathcal{L}\{S\}, \tag{31}$$
$$S^{\phi_4}\mathcal{L}\{Q\} = S^{\phi_4-1}Q(0) - \alpha_2\mathcal{L}\{SQ\} - \mu\mathcal{L}\{Q\} + \gamma(1-\delta)\mathcal{L}\{S\}, \tag{32}$$
$$S^{\phi_5}\mathcal{L}\{L\} = S^{\phi_5-1}L(0) + \delta\gamma\mathcal{L}\{S\} - \mu\mathcal{L}\{L\}, \tag{33}$$

by using the initial conditions (7), we get

$$\mathcal{L}\{P\} = \frac{n_1}{S} + \frac{\Lambda}{S^{\phi_1+1}} - \frac{\beta}{S^{\phi_1}}\mathcal{L}\{PS\} - \frac{\mu}{S^{\phi_1}}\mathcal{L}\{P\}, \tag{34}$$
$$\mathcal{L}\{O\} = \frac{n_2}{S} + \frac{\beta}{S^{\phi_2}}\mathcal{L}\{PS\} - \frac{\alpha_1+\mu}{S^{\phi_2}}\mathcal{L}\{O\}, \tag{35}$$
$$\mathcal{L}\{S\} = \frac{n_3}{S} + \frac{\alpha_1}{S^{\phi_3}}\mathcal{L}\{O\} + \frac{\alpha_2}{S^{\phi_3}}\mathcal{L}\{SQ\} - \frac{\mu+\gamma}{S^{\phi_3}}\mathcal{L}\{S\}, \tag{36}$$
$$\mathcal{L}\{Q\} = \frac{n_4}{S} - \frac{\alpha_2}{S^{\phi_4}}\mathcal{L}\{SQ\} - \frac{\mu}{S^{\phi_4}}\mathcal{L}\{Q\} + \frac{\gamma(1-\delta)}{S^{\phi_4}}\mathcal{L}\{S\}, \tag{37}$$
$$\mathcal{L}\{L\} = \frac{n_5}{S} + \frac{\delta\gamma}{S^{\phi_5}}\mathcal{L}\{S\} - \frac{\mu}{S^{\phi_5}}\mathcal{L}\{L\}. \tag{38}$$

It should be assumed that method gives the solution as an infinite series

$$P = \sum_{k=0}^{\infty} P_k,\ O = \sum_{k=0}^{\infty} O_k,\ S = \sum_{k=0}^{\infty} S_k,\ Q = \sum_{k=0}^{\infty} Q_k,\ L = \sum_{k=0}^{\infty} L_k, \tag{39}$$

The nonlinearity PS and SQ can be written as

$$PS = \sum_{k=0}^{\infty} A_k,\quad SQ = \sum_{k=0}^{\infty} B_k, \tag{40}$$

where A_k and B_k is called the Adomian polynomials given as

$$A_k = \frac{1}{k!}\frac{d^k}{\lambda^k}\left[\sum_{j=0}^{k}\lambda^j P_j \sum_{j=0}^{k}\lambda^j S_j\right]|_{\lambda=0},\quad B_k = \frac{1}{k!}\frac{d^k}{\lambda^k}\left[\sum_{j=0}^{k}\lambda^j Q_j \sum_{j=0}^{k}\lambda^j S_j\right]|_{\lambda=0}. \tag{41}$$

Substitute equations (39) and (40) in (34–38), we have the followings results

$$\mathcal{L}\{P_0\} = \frac{n_1}{S} + \frac{\Lambda}{S^{\phi_1+1}},\ \mathcal{L}\{O_0\} = \frac{n_2}{S},\ \mathcal{L}\{S_0\} = \frac{n_3}{S}, \tag{42}$$

$$\mathcal{L}\{Q_0\} = \frac{n_4}{S}, \ \mathcal{L}\{L_0\} = \frac{n_5}{S}, \tag{43}$$

similarly, we have

$$\mathcal{L}\{P_1\} = -\frac{\beta}{S^{\phi_2}}\mathcal{L}\{A_0\}\frac{\alpha_1+\mu}{S^{\phi_2}}\mathcal{L}\{O_0\} + \cdots, \tag{44}$$

$$\mathcal{L}\{P_{k+1}\} = -\frac{\beta}{S^{\phi_2}}\mathcal{L}\{A_k\} - \frac{\alpha_1+\mu}{S^{\phi_2}}\mathcal{L}\{O_k\}, \tag{45}$$

$$\mathcal{L}\{O_1\} = \frac{\beta}{S^{\phi_2}}\mathcal{L}\{A_0\} - \frac{\alpha_1+\mu}{S^{\phi_2}}\mathcal{L}\{O_0\} + \cdots, \tag{46}$$

$$\mathcal{L}\{O_{k+1}\} = \frac{\beta}{S^{\phi_2}}\mathcal{L}\{A_k\} - \frac{\alpha_1+\mu}{S^{\phi_2}}\mathcal{L}\{O_k\}, \tag{47}$$

$$\mathcal{L}\{S_1\} = \frac{\alpha_1}{S^{\phi_3}}\mathcal{L}\{O_0\} + \frac{\alpha_2}{S^{\phi_3}}\mathcal{L}\{B_0\} - \frac{\mu+\gamma}{S^{\phi_3}}\mathcal{L}\{S_0\} + \cdots, \tag{48}$$

$$\mathcal{L}\{S_{k+1}\} = \frac{\alpha_1}{S^{\phi_3}}\mathcal{L}\{O_k\} + \frac{\alpha_2}{S^{\phi_3}}\mathcal{L}\{B_k\} - \frac{\mu+\gamma}{S^{\phi_3}}\mathcal{L}\{S_k\}, \tag{49}$$

$$\mathcal{L}\{Q_1\} = -\frac{\alpha_2}{S^{\phi_4}}\mathcal{L}\{B_0\} - \frac{\mu}{S^{\phi_4}}\mathcal{L}\{Q_0\} + \frac{\gamma(1-\delta)}{S^{\phi_4}}\mathcal{L}\{S_0\} + \cdots, \tag{50}$$

$$\mathcal{L}\{Q_{k+1}\} = -\frac{\alpha_2}{S^{\phi_4}}\mathcal{L}\{B_k\} - \frac{\mu}{S^{\phi_4}}\mathcal{L}\{Q_k\} + \frac{\gamma(1-\delta)}{S^{\phi_4}}\mathcal{L}\{S_k\}, \tag{51}$$

$$\mathcal{L}\{L_1\} = \frac{\delta\gamma}{S^{\phi_5}}\mathcal{L}\{S_0\} - \frac{\mu}{S^{\phi_5}}\mathcal{L}\{L_0\} + \cdots, \tag{52}$$

$$\mathcal{L}\{L_{k+1}\} = \frac{\delta\gamma}{S^{\phi_5}}\mathcal{L}\{S_k\}\frac{\mu}{S^{\phi_5}}\mathcal{L}\{L_k\}, \tag{53}$$

The purpose of the work is to analyze the mathematical behavior of the solution $P(t)$, $O(t)$, $S(t)$, $Q(t)$, $L(t)$ for the different values of ϕ. By applying the inverse Laplace transform to both sides of the Eq. (42), we get the values of P_0, O_0, S_0, Q_0, L_0 and used for further process. Putting the values of P_0, O_0, S_0, Q_0, L_0 and A_0, B_0 into the equations (43–47)and get the values of P_1, O_1, S_1, Q_1, L_1. Similarly, we find the remaining term $P_2, P_3, P_4, \ldots, O_2, O_3, O_4, \ldots, S_2, S_3, S_4, \ldots, Q_2, Q_3, Q_4, \ldots$ and $L_2, L_3, L_4, \ldots$ in the same manners. We have the series solution in the form (Table 1)

$$P(t) = P_0 + P_1 + P_2 + P_3 + P_4 + \cdots, \tag{54}$$

$$O(t) = O_0 + O_1 + O_2 + O_3 + O_4 + \cdots, \tag{55}$$

$$S(t) = S_0 + S_1 + S_2 + S_3 + S_4 + \cdots, \tag{56}$$

$$Q(t) = Q_0 + Q_1 + Q_2 + Q_3 + Q_4 + \cdots \tag{57}$$

$$L(t) = L_0 + L_1 + L_2 + L_3 + L_4 + \cdots, \tag{58}$$

We have computed first three terms of the series by using the LADM for the equations (6)-(10) as

$$\mathcal{L}\{P_0\} = \frac{n_1}{S} + \frac{\Lambda}{S^{\phi_1+1}}, \ \mathcal{L}\{O_0\} = \frac{n_2}{S}, \ \mathcal{L}\{S_0\} = \frac{n_3}{S}, \ \mathcal{L}\{Q_0\} = \frac{n_4}{S}, \ \mathcal{L}\{L_0\} = \frac{n_5}{S},$$

by using Laplace inverse property with initial conditions $n_1 = 40$, $n_2 = 10$, $n_3 = 20$, $n_4 = 10$ and $n_5 = 5$, we have

$$P_0 = 40 + \frac{t_1^{\phi}}{\phi_1!}, \quad O_0 = 10, \quad S_0 = 20, \quad Q_0 = 10, \quad L_0 = 5,$$

Table 1 Values of physical parameters used in fractional order smoking model

Parameter	Value	Unit
Λ	1	1/time
β	0.14	1/time
μ	0.05	1/time
δ	0.1	1/time
γ	0.8	1/time
α_1	0.002	1/time
α_2	0.0025	1/time

$$P_1 = \frac{-114t^{\phi_1}}{\phi_1!} - \frac{2.8t^{2\phi_1}}{2\phi_1!},\ O_1 = \frac{111.48t^{\phi_2}}{\phi_2!} + 2.8\frac{t^{\phi_1+\phi_2}}{(\phi_1+\phi_2)!},$$
$$S_1 = -16.48\frac{t^{\phi_3}}{\phi_3!},\ Q_1 = 13.4\frac{t^{\phi_4}}{\phi_4!},\ L_1 = 15.75\frac{t^{\phi_5}}{\phi_5!},$$
$$P_2 = 324.9\frac{t^{2\phi_1}}{2\phi_1!} + 8.1225\frac{t^{3\phi_1}}{3\phi_1!} + 92.288\frac{t^{\phi_1+\phi_3}}{(\phi_1+\phi_3)!} + 2.3072\frac{(\phi_1+\phi_3)!t^{2\phi_1+\phi_3}}{\phi_1!\phi_3!\,(2\phi_1+\phi_3)!},$$
$$O_2 = -319.2\frac{t^{\phi_1+\phi_2}}{(\phi_1+\phi_2)!} - 7.98\frac{t^{2\phi_1+\phi_2}}{(2\phi_1+\alpha_2)!} - 92.88\frac{t^{\phi_2+\phi_3}}{(\phi_2+\phi_3)!}$$
$$-2.3072\frac{(\phi_1+\phi_3)!t^{\phi_1+\phi_2+\phi_3}}{\phi_1!\phi_3!\,(\phi_1+\phi_2+\phi_3)!}, -5.797\frac{t^{2\phi_2}}{(2\phi_2)!} - 0.1456\frac{t^{\phi_1+2\phi_2}}{(\phi_1+2\phi_2)!},$$
$$S_2 = 0.2296\frac{t^{\phi_2+\phi_3}}{(\phi_2+\phi_3)!} + 0.0056\frac{t^{\phi_1+\phi_2+\phi_3}}{(\phi_1+\phi_2+\phi_3)!} + 13.596\frac{t^{2\phi_3}}{(2\phi_3)!} + 0.0067\frac{t^{\phi_3+\phi_4}}{(\phi_3+\phi_4)!},$$
$$Q_2 = -16.4536\frac{t^{\phi_3+\phi_4}}{(\phi_3+\phi_4)!} - 0.6767\frac{t^{2\phi_4}}{2\phi_4!},$$
$$L_2 = -0.7875\frac{t^{2\phi_5}}{2\phi_5!} - 13.184\frac{t^{\phi_3+\phi_5}}{(\phi_3+\phi_5)!},$$

The solution of the fractional model in series form is written as

$$P(t) = 40 - 113\frac{t^{\phi_1}}{\phi_1!} + 322.1\frac{t^{2\phi_1}}{2\phi_1!} + 8.1225\frac{t^{3\phi_1}}{3\phi_1!} + 92.288\frac{t^{\phi_1+\phi_3}}{(\phi_1+\phi_3)!}$$
$$+ 2.3072\frac{(\phi_1+\phi_3)!t^{2\phi_1+\phi_3}}{\phi_1!\phi_3!\,(2\phi_1+\phi_3)!}, \tag{59}$$
$$O(t) = 10 + \frac{111.48t^{\phi_2}}{\phi_2!} - 316.4\frac{t^{\phi_1+\phi_2}}{(\phi_1\phi_2)!} - 7.98\frac{t^{2\phi_1+\phi_2}}{(2\phi_1+\alpha_2)!} - 92.88\frac{t^{\phi_2+\phi_3}}{(\phi_2+\phi_3)!}$$
$$- 2.3072\frac{(\phi_1+\phi_3)!t^{\phi_1+\phi_2+\phi_3}}{\phi_1!\phi_3!\,(\phi_1+\phi_2+\phi_3)!}, \tag{60}$$
$$S(t) = 20 - 16.48\frac{t^{\phi_3}}{\phi_3!} + 0.2296\frac{t^{\phi_2+\phi_3}}{(\phi_2+\phi_3)!} - 0.0056\frac{t^{\phi_1+\phi_2+\phi_3}}{(\phi_1+\phi_2+\phi_3)!} + 13.596\frac{t^{2\phi_3}}{(2\phi_3)!}$$
$$+ 0.0067\frac{t^{\phi_3+\phi_4}}{(\phi_3+\phi_4)!}, \tag{61}$$
$$Q(t) = 10 + 13.4\frac{t^{\phi_4}}{\phi_4!} - 16.4536\frac{t^{\phi_3+\phi_4}}{(\phi_3+\phi_4)!} - 0.6767\frac{t^{2\phi_4}}{2\phi_4!}, \tag{62}$$

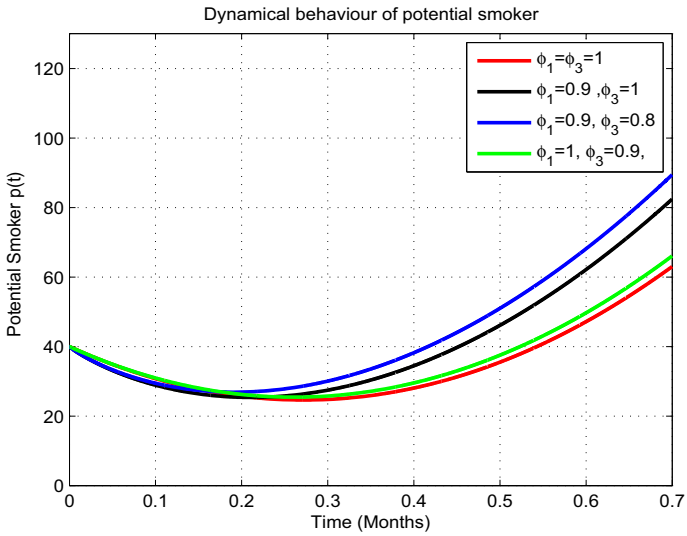

Fig. 1 Numerical solutions for potential smokers in a time t (months)

$$L(t) = 5 + 1.35\frac{t^{\phi_5}}{\phi_5!} - 0.7875\frac{t^{2\phi_5}}{2\phi_5!} - 13.184\frac{t^{\phi_3+\phi_5}}{(\phi_3+\phi_5)!}, \tag{63}$$

In particular, the solution of the model with fractional derivatives for $\phi_1 = \phi_2 = \phi_3 = \phi_4 = \phi_5 = 1$, is given by

$$P(t) = 40 - 113t + 207.169t^2 + 2.1228t^3, \tag{64}$$
$$O(t) = 10 + 111.48t - 207.2425t^2 - 3.453t^3, \tag{65}$$
$$S(t) = 20 - 16.48t + 6.9129t^2 + 0.0009t^3, \tag{66}$$
$$Q(t) = 10 + 13.4t - 8.5652t^2, \tag{67}$$
$$L(t) = 5 + 15.75t - 6.9858t^2. \tag{68}$$

Numerical Results and Discussion

We conclude several numerical simulations varying the value of fractional parameter. These simulations reveals that a change of the value affects the dynamics of the epidemic. In Figs. (1, 2, 3, 4, 5), we can observe that the epidemic peak is wider and higher for true equilibrium point. Similarly, it is true for the endemic equilibrium points (Tables 2, 3, 4, 5, 6).

The mathematical analysis of epidemic smoking model with non-linear incidence has been presented. To observe the effects of the parameters using in this dynamics smoking model given in equation (6), several numerical simulations are presented by varying the value of fractional parameter ϕ. These simulations reveals the effect for integer order values as well as non-integer values of ϕ. The non-integer order has minor affect in the dynamics of the epidemic as shown in Figs. (1, 2, 3, 4, 5). This also shows that the technique applied for fractional differential equations gives better approximation for different fractional model. We employed integer, fractional as well as mixed integer and fractional values of ϕ for potential smoker (non smoker) and it can be observed that the non smoker increases at fractional values

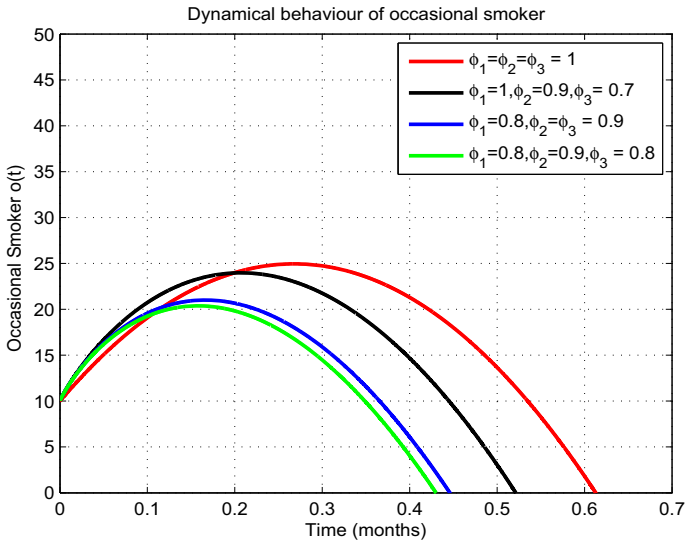

Fig. 2 Numerical solutions for occasional smokers in a time t (months)

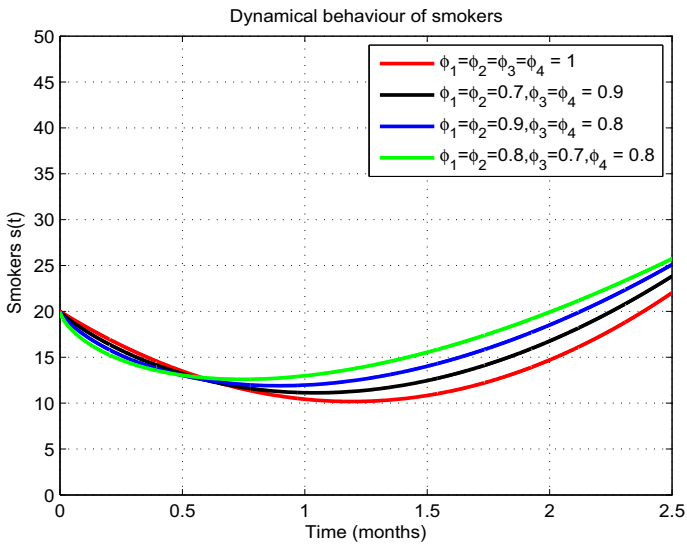

Fig. 3 Numerical solutions for smokers in a time t (months)

of ϕ. For the equilibrium point one of the infected component is non zero i.e $S(t)$ and its converging behavior can be seen for integer as well as fractional values of ϕ, while the other infected component converges to zero. It can also be observed that the infected rate decreases as decreasing the fractional values of ϕ.

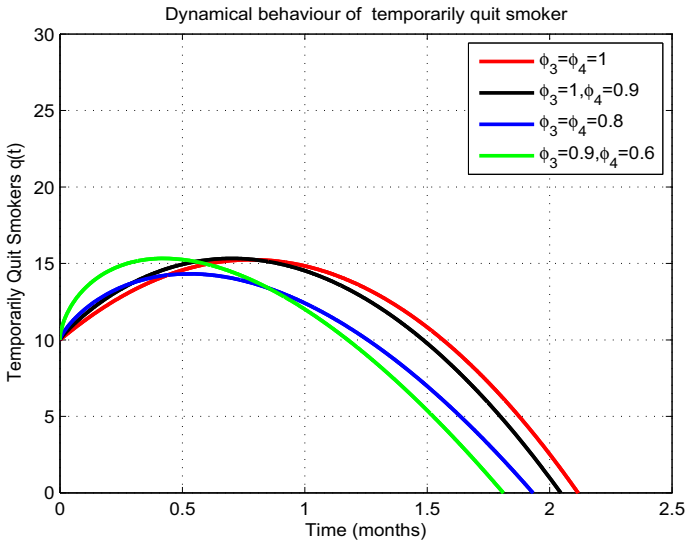

Fig. 4 Numerical solutions for temporarily quit smokers in a time t (months)

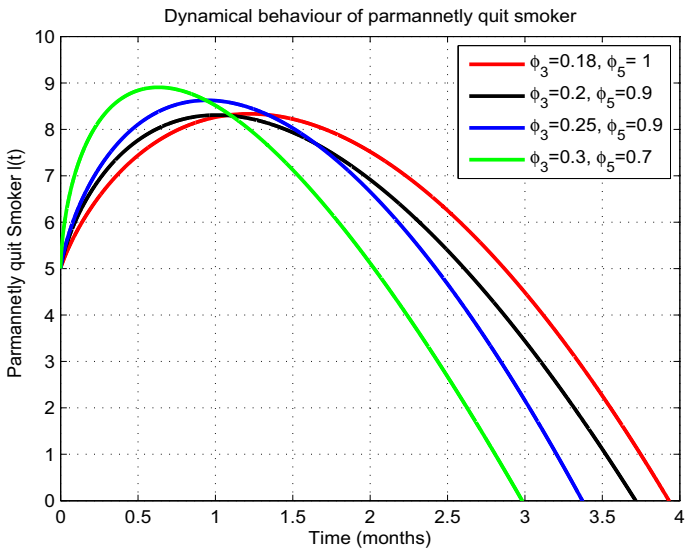

Fig. 5 Numerical solutions for permanently quit smokers in a time t (months)

Convergence Analysis

The obtained series solution is rapidly convergent and also converges uniformly to the exact solution. We use the classical techniques to verify the convergence of the series of model. We check the condition of convergence of the method by using the idea of the following theorem [24].

Table 2 Table of potential smoker $P(t)$ at different values of $\phi_i, i = 1, 2, 3, 4, 5$

t	$\phi_1 = 1$	$\phi_1 = 0.8$	$\phi_1 = 0.9$	$\phi_1 = 1$	$\phi_1 = 1$
	$\phi 2 = 1$	$\phi_2 = 0.8$	$\phi_2 = 0.9$	$\phi_2 = 1$	$\phi_2 = 1$
	$\phi_3 = 1$	$\phi_3 = 0.8$	$\phi_3 = 0.9$	$\phi_3 = 1$	$\phi_3 = 0.8$
	$\phi_4 = 1$	$\phi_4 = 0.8$	$\phi_4 = 0.9$	$\phi_4 = 0.6$	$\phi_4 = 0.8$
	$\phi_5 = 1$	$\alpha_5 = 0.8$	$\phi_5 = 0.9$	$\phi_5 = 0.6$	$\phi_5 = 0.8$
0	40	40	40	40	40
0.1	30.7738	28.6749	29.132	30.7738	31.1855
0.2	25.7037	28.6749	26.0772	25.7037	26.8996
0.3	24.8025	36.1378	28.6572	24.8025	26.9629
0.4	28.0829	49.0558	36.24	28.0829	31.2978
0.5	35.5576	66.6838	48.4674	35.5576	39.8609

Table 3 Table of occasional smoker $O(t)$ at different values of $\phi_i, i = 1, 2, 3, 4, 5$

t	$\phi_1 = 1$	$\phi_1 = 0.8$	$\phi_1 = 0.9$	$\phi_1 = 1$	$\phi_1 = 1$
	$\phi 2 = 1$	$\phi_2 = 0.8$	$\phi_2 = 0.9$	$\phi_2 = 1$	$\phi_2 = 1$
	$\phi_3 = 1$	$\phi_3 = 0.8$	$\phi_3 = 0.9$	$\phi_3 = 1$	$\phi_3 = 0.8$
	$\phi_4 = 1$	$\phi_4 = 0.8$	$\phi_4 = 0.9$	$\phi_4 = 0.6$	$\phi_4 = 0.8$
	$\phi_5 = 1$	$\alpha_5 = 0.8$	$\phi_5 = 0.9$	$\phi_5 = 0.6$	$\phi_5 = 0.8$
0	10	10	10	10	10
0.1	19.0721	21.6711	20.6677	19.0721	17.2388
0.2	23.9787	20.8669	23.5467	23.9787	18.6613
0.3	24.6999	13.2241	20.798	24.6989	15.1152
0.4	21.2122	6.1356	13.0504	21.2122	6.9820
0.5	13.4978	− 17.6563	0.6605	13.4978	− 5.4949

Table 4 Table of smoker $S(t)$ at different values of $\phi_i, i = 1, 2, 3, 4, 5$

t	$\phi_1 = 1$	$\phi_1 = 0.8$	$\phi_1 = 0.9$	$\phi_1 = 1$	$\phi_1 = 1$
	$\phi 2 = 1$	$\phi_2 = 0.8$	$\phi_2 = 0.9$	$\phi_2 = 1$	$\phi_2 = 1$
	$\phi_3 = 1$	$\phi_3 = 0.8$	$\phi_3 = 0.9$	$\phi_3 = 1$	$\phi_3 = 0.8$
	$\phi_4 = 1$	$\phi_4 = 0.8$	$\phi_4 = 0.9$	$\phi_4 = 0.6$	$\phi_4 = 0.8$
	$\phi_5 = 1$	$\alpha_5 = 0.8$	$\phi_5 = 0.9$	$\phi_5 = 0.6$	$\phi_5 = 0.8$
0	20	20	20	20	20
0.1	18.1142	17.4386	17.9736	18.4212	17.4368
0.2	16.9805	15.8539	16.4298	16.9807	15.8493
0.3	15.6782	14.6555	15.1463	15.6785	14.6479
0.4	14.5141	13.7315	14.0735	14.5147	13.7209
0.5	13.4883	13.0281	13.1863	13.4890	13.0145

Theorem 3 *Let Y be a Banach space and $F : Y \to Y$ be a contractive nonlinear operator then there exit $y, y' \in Y$, $||F(y) - F(y')|| \leq k||y - y'||$, $0 < k < 1$. Then F has a unique point y such that $Fy = y$, where $y = (P, O, S, Q, L)$. The series solution of model by using*

Table 5 Table of temporarily quit smoker $Q(t)$ at different values of $\phi_i, i = 1, 2, 3, 4, 5$

t	$\phi_1 = 1$	$\phi_1 = 0.8$	$\phi_1 = 0.9$	$\phi_1 = 1$	$\phi_1 = 1$
	$\phi 2 = 1$	$\phi_2 = 0.8$	$\phi_2 = 0.9$	$\phi_2 = 1$	$\phi_2 = 1$
	$\phi_3 = 1$	$\phi_3 = 0.8$	$\phi_3 = 0.9$	$\phi_3 = 1$	$\phi_3 = 0.8$
	$\phi_4 = 1$	$\phi_4 = 0.8$	$\phi_4 = 0.9$	$\phi_4 = 0.6$	$\phi_4 = 0.8$
	$\phi_5 = 1$	$\alpha_5 = 0.8$	$\phi_5 = 0.9$	$\phi_5 = 0.6$	$\phi_5 = 0.8$
0	10	10	10	10	10
0.1	11.2543	11.9806	11.592	13.4392	11.9792
0.2	12.3374	13.0618	12.7091	14.7445	13.6567
0.3	13.2429	13.7537	13.5444	15.4611	13.7456
0.4	13.9896	14.1591	14.1439	15.7933	14.1463
0.5	14.5587	14.3278	14.5317	15.8305	14.3105

Table 6 Table of permanently quit smoker $L(t)$ at different values of $\phi_i, i = 1, 2, 3, 4, 5$

t	$\phi_1 = 1$	$\phi_1 = 0.8$	$\phi_1 = 0.9$	$\phi_1 = 1$	$\phi_1 = 1$
	$\phi 2 = 1$	$\phi_2 = 0.8$	$\phi_2 = 0.9$	$\phi_2 = 1$	$\phi_2 = 1$
	$\phi_3 = 1$	$\phi_3 = 0.8$	$\phi_3 = 0.9$	$\phi_3 = 1$	$\phi_3 = 0.8$
	$\phi_4 = 1$	$\phi_4 = 0.8$	$\phi_4 = 0.9$	$\phi_4 = 0.6$	$\phi_4 = 0.8$
	$\phi_5 = 1$	$\alpha_5 = 0.8$	$\phi_5 = 0.9$	$\phi_5 = 0.6$	$\phi_5 = 0.8$
0	5	5	5	5	5
0.1	6.6449	7.4346	6.9295	9.1508	7.4346
0.2	8.4294	8.9221	8.3871	10.9049	8.9222
0.3	10.3537	10.0305	9.5870	12.8045	10.0305
0.4	12.4177	10.8685	10.5772	12.8045	10.8685
0.5	14.6215	11.4884	11.3822	13.2751	11.4884

ADM technique is given as:

$$y_n = Ty_{n-1}, \quad y_{n-1} = \sum_{j=1}^{n-1} y_j, \; n = 1, 2, 3, \ldots,$$

and suppose that $y_0 \in B_r(y)$ *where* $B_r(y) = \{y' \in Y : ||y' - y|| < r\}$, *then we get*

(i) $y_n \in B_r(x)$;
(ii) $\lim_{n\to\infty} y_n = y$.

Proof For (i) by using mathematical induction for $n = 1$, we obtained

$$||y_0 - y|| = ||F(y_0) - F(x)|| \leq K||y_0 - y||.$$

suppose that the statement is true for $m - 1$, then

$$||y_0 - y|| \leq K^{m-1}||y_0 - y||.$$

we get

$$||y_m - y|| = ||F(y_{m-1}) - F(x)|| \leq K||y_{m-1} - y|| \leq K^m||y_0 - y||.$$

$$||y_m - y|| \leq K^n||y_0 - y|| \leq K^n r \leq r.$$

which implies that $y_n \in B_r(x)$.

(ii) Since$||y_m - y|| \leq K^n||y_0 - y||$ and $\lim_{n\to\infty} k^n = 0$, therefore, we have the $\lim_{n\to\infty} ||y_n - y|| = 0, \Longrightarrow \lim_{n\to\infty} y_n = y$. □

Conclusion

Human beings are facing harmful diseases including throat, mouth, stomach, cervix, breast, pancreas cancer and many others caused by smoking. In this study smoking epidemic model has been investigated. It is important to note that Laplace Adomian decomposition method for mathematical models based on system of fractional order differential equations is more powerful approach to compute the convergent solutions. The convergence analysis is also provided to demonstrate the efficiency of the method. The constructed series by Laplace Adomian decomposition method for smoking model show a good agreement to control the bad impact of smoking for different time period and to eradicate a death killer factor in the world. we introduced the stability analysis theory and sensitivity analysis of mathematical epidemic models in the nonlinear system which represent both the local and global behavior of smoking dynamics. Finally, we estimated the parameter that characterize the behavior of disease and present numerical simulations. The disease free equilibrium state is carried out by presenting $\mathrm{Re}(\lambda) < 0$, i.e. the population is viable.

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